

1. (Hyman 4.1) The following table shows how the marginal benefit of a service varies for four consumers:

Marginal Benefit (in Dollars)				
	CONSUMERS			
QUANTITY	ALICE	BEN	CAROLYN	DON
1	1000	800	600	400
2	800	600	400	200
3	600	400	200	100
4	400	200	100	50

- Suppose the service is a pure private good and is sold in a competitive market with the only buyers being the four people whose marginal benefits are shown in the table. If the market price of the product is \$400, what is the quantity demanded?
- Suppose the service is a pure public good with the only consumers being the four people whose marginal benefits are shown in the table. What is the marginal social benefit of two units of the service?
- If the marginal social cost of the good is \$2,000, what is the efficient output assuming that it is a pure private good?
- If the marginal social cost of the good is \$2,000, what is the efficient output assuming it is a pure public good?

ANS

a. 10

b. \$2,000

c. zero

d. 2

2. (Hyman 4.2) Suppose the marginal cost of a pure public good increases as more is purchased by a community. Prove that the Lindahl equilibrium will result in a budget surplus at the efficient annual output of the pure public good.

ANS

The sum of the marginal benefits must equal the marginal cost in the Lindahl equilibrium. When MC is increasing, $MC > AC$ for any given quantity supplied. At the

efficient level of output, the sum of the contributions will be greater than AC.

Because contributions per unit exceed cost per unit, the total revenue collected will exceed the cost of making the good available, which is $AC(Q)$.

Thus, there is a budget surplus.

3. (Hyman 4.3) Suppose the services of a road are subject to congestion after 50,000 vehicles per hour enter the road. Assume that it is feasible to price road services on an hourly basis. Use a graph like that drawn in Figure 4.2 to show how the services of the road should be priced per hour when fewer than and more than 50,000 vehicles per hour are expected so as to achieve efficiency.

ANS

At less than 50,000 vehicles per hour, no toll is required to achieve efficiency. When traffic rises above 50,000 vehicles per hour, the toll should be set at the marginal congestion cost.

4. (Hyman 4.4) The following table shows how the marginal benefit enjoyed by John, Mary, Loren, and all other consumers of outdoor rock concerts varies with the number made available by a city government per summer.

Marginal Benefit of Number of Rock Concerts per Consumer (in Dollars)

	NUMBER OF CONCERTS			
CONSUMERS	1	2	3	4
John	150	125	100	75
Mary	125	100	75	50
Loren	100	75	50	25
All Others	600	400	200	100

- Derive the demand curve for rock concerts assuming that it is a pure public good.
- If the marginal cost of producing rock concerts is \$1,000 no matter how many are produced, then what is the efficient number of concerts to have each summer? What would be the efficient number of concerts to produce if the marginal cost of production were \$425 instead of \$1,000?

ANS

- The demand curve shows how the sum of the marginal benefits of all consumers varies with the number of concerts.

Number of Concerts	MB
1	\$975

2	\$700
3	\$425
4	\$250

- b. At a marginal cost of \$1,000, it is efficient not to hold any outdoor rock concerts at all. At a marginal cost of \$425, the efficient number of concerts per summer is three.

5. (Hyman 4.5) Suppose the marginal cost of producing rock concerts is only \$250 per concert no matter how many are produced. Use the data from the previous question to calculate the efficient number of concerts. If a Lindahl scheme is used to finance the concerts, what prices of admission should be charged to John, Loren, and Mary?

ANS

The efficient number of concerts would be four; John would be charged \$75, and Mary would be charged \$50, while Loren would pay \$25 for each concert.

6. (Rosen 4.3) Tarzan and Jane live alone in the jungle and have trained Cheetah both to patrol the perimeter of their clearing and to harvest tropical fruits. Cheetah can collect 3 pounds of fruit an hour and currently spends 6 hours patrolling, 8 hours picking, and 10 hours sleeping.

- What are the public and private goods in this example?
- If Tarzan and Jane are each currently willing to give up one hour of patrol for 2 pounds of fruit, is the current allocation of Cheetah's time Pareto efficient? Should he patrol more or less?

ANS

We assume that Cheetah's utility does not enter the social welfare function; hence, her allocation of labor supply across activities does not matter.

- The public good is patrol; the private good is fruit.
- Recall that efficiency requires $MRS^{\text{TARZAN}} + MRS^{\text{JANE}} = \text{MRT}$. $MRS^{\text{TARZAN}} = MRS^{\text{JANE}} = 2$. But $\text{MRT} = 3$. Therefore, $MRS^{\text{TARZAN}} + MRS^{\text{JANE}} < \text{MRT}$. To achieve an efficient allocation, Cheetah should patrol more.

